

TINDAL, Australia

WMO: 941310

Lat: 14.5228S

Lon: 132.3825E

Elev: 443

StdP: 14.46

Time Zone: 9.50 (AUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	49.7	52.1	19.7	15.1	79.6	24.9	19.2	76.1	18.5	77.1	16.9	77.9	5.5	130	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	21.4	102.2	70.8	100.4	71.2	98.7	71.7	83.8	88.7	80.3	89.3	79.2	88.1	8.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.7	173.4	85.5	78.6	150.9	83.5	77.2	143.9	82.1	48.5	90.5	44.5	89.4	43.3	88.9	99.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	15.3	13.8	43.5	104.6	3.4	1.8	41.1	105.9	39.1	107.0	37.2	108.0	34.7	109.3		
(5)			38.9	88.2	2.5	6.3	37.1	92.8	35.6	96.5	34.2	100.0	32.4	104.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	84.1	84.0	83.2	80.8	76.2	71.6	71.4	73.7	81.6	86.8	87.6	86.2	(6)
(7)		DBStd	6.84	2.91	2.58	2.51	3.28	4.70	5.23	4.42	4.57	4.50	3.42	3.26	3.29	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	16	0	0	0	0	1	9	5	1	0	0	0	0	(9)
(10)		CDD50	11163	1057	951	1030	925	813	648	663	735	949	1141	1129	1122	(10)
(11)		CDD65	5704	592	531	565	475	348	207	203	272	499	676	679	657	(11)
(12)		CDH74	70300	6562	5832	6235	5459	3868	2321	2531	3651	6814	9678	9257	8092	(12)
(13)		CDH80	36230	3017	2692	2981	2781	1789	895	1000	1729	3857	5796	5425	4268	(13)
(14)	Wind	WSAvg	6.4	6.0	5.7	5.4	6.2	6.9	7.2	7.2	6.8	6.4	6.5	6.0	5.9	(14)
(15)	Precipitation	PrecAvg	43.3	10.5	8.3	7.0	1.5	0.1	0.0	0.0	0.1	0.2	1.2	3.6	10.6	(15)
(16)		PrecMax	62.6	26.7	20.2	16.4	7.6	1.0	0.3	0.1	1.2	2.2	4.4	10.2	22.3	(16)
(17)		PrecMin	24.6	3.8	1.4	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	4.1	(17)
(18)		PrecStd	10.2	5.5	5.2	4.8	2.1	0.2	0.1	0.0	0.2	0.5	1.5	2.3	5.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.8	98.8	98.8	97.1	94.9	92.0	92.2	95.3	100.3	103.9	104.3	104.1	(19)
(20)			MCWB	77.2	74.6	74.7	72.3	71.1	68.9	67.7	67.0	68.2	69.9	70.0	72.8	(20)
(21)		2%	DB	96.6	96.7	96.5	95.2	92.9	89.8	89.8	93.2	98.5	102.0	102.3	100.4	(21)
(22)			MCWB	77.2	75.9	74.7	72.0	70.4	67.9	65.9	65.9	67.7	69.8	70.4	75.3	(22)
(23)		5%	DB	94.7	94.8	94.5	93.6	91.1	87.9	87.9	91.1	96.9	100.3	100.4	98.2	(23)
(24)			MCWB	77.7	76.6	74.7	71.6	69.8	67.0	64.6	64.5	67.0	69.8	71.2	76.2	(24)
(25)		10%	DB	92.8	92.8	92.8	91.9	89.3	85.7	86.0	88.9	95.1	98.5	98.4	95.2	(25)
(26)			MCWB	77.9	77.2	74.7	71.4	68.9	65.4	63.4	63.4	66.6	69.6	72.6	76.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.1	82.9	87.7	83.0	82.6	73.7	72.8	72.4	79.2	91.6	89.5	90.5	(27)
(28)			MCDWB	92.6	88.9	88.7	84.7	85.3	83.1	84.7	85.3	90.4	94.9	89.5	92.0	(28)
(29)		2%	WB	80.6	80.3	80.2	77.3	75.4	70.9	69.9	70.1	74.5	78.4	78.8	80.0	(29)
(30)			MCDWB	90.2	88.9	88.3	85.1	85.3	84.7	83.8	85.4	85.7	87.5	89.3	89.8	(30)
(31)		5%	WB	79.6	79.3	78.8	76.3	73.6	69.3	67.9	68.4	72.9	75.6	77.4	79.0	(31)
(32)			MCDWB	88.7	87.8	86.7	84.4	84.6	83.2	82.5	85.0	86.3	87.2	88.5	88.9	(32)
(33)		10%	WB	78.8	78.6	77.9	75.4	72.0	67.7	66.1	66.6	71.7	74.6	76.5	78.2	(33)
(34)			MCDWB	87.5	86.8	85.5	83.7	83.3	81.4	81.2	83.6	86.6	86.8	87.7	87.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.3	16.7	18.5	22.9	25.5	27.1	29.1	30.3	27.7	23.7	21.4	18.6	(35)
(36)			MCDBR	19.6	20.4	22.3	25.0	25.4	26.8	29.7	31.2	27.7	25.7	23.8	22.0	(36)
(37)			MCWBR	6.1	6.1	6.5	7.3	8.4	9.4	10.9	10.8	9.6	9.9	8.2	6.3	(37)
(38)		5% WB	MCDBR	17.0	16.6	17.0	18.8	21.8	24.7	26.4	26.1	22.9	21.7	20.2	18.2	(38)
(39)			MCWBR	6.1	6.0	6.2	6.6	9.1	9.9	11.1	10.4	9.2	9.2	6.6	6.1	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.407	0.403	0.390	0.369	0.356	0.332	0.327	0.342	0.406	0.435	0.450	0.425	(40)
(41)	taud		2.496	2.503	2.535	2.562	2.537	2.552	2.559	2.488	2.325	2.277	2.292	2.421	(41)	
(42)	Ebn at Noon		298	296	294	289	283	285	289	293	284	283	283	292	(42)	
(43)	Edh at Noon		37	36	34	31	30	29	29	33	42	45	45	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1914	1941	1912	1885	1713	1633	1729	1961	2143	2221	2141	2028	(44)	
(45)		RadStd	170	228	202	142	98	78	52	61	108	120	77	134	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page